

TEL-OS: INFERENCE-ONLY REPRESENTATION GOVERNANCE ELIMINATES ADVERSARIAL JAIL-BREAK OPTIMIZATION

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ABSTRACT

Optimized adversarial attacks—genetic suffix optimization, semantic inversion, stacked identity-hijacking, and autonomous multi-turn LRM agents—reduce the safety of RLHF-aligned LLMs to near zero: frontier models suffer 97.14% Attack Success Rate (ASR) against LRM autonomous agents (Hagendorff et al., 2026), and existing inference-time defenses fail to close this gap without weight modification or prohibitive overhead. We present **TEL-OS v3.0**, an inference-only governance framework that closes this gap via *adaptive per-layer refusal steering*: per-layer refusal direction extraction (OBLITERATUS), dual-layer OR-logic detection at architecturally consistent depths ($\sim 40\%$ and $\sim 70\%$), and urgency-scaled geometry-preserving steering with zero weight modification.

On Llama-3.1-8B-Instruct, TEL-OS achieves **0.00% ASR** against every optimized attack paradigm evaluated: LRM autonomous multi-turn attacks are blocked across 200 independent trials (4 frontier attackers: DeepSeek-R1, GPT-4o, Qwen3-235B, Gemini 2.5 Flash), 95% CI [0.00%, 1.83%], model-as-a-judge evaluation; AutoDAN-Turbo genetic optimization produces `best_loss` = 0.5 from step 1 with no improvement across 50 behaviors; FlipAttack semantic inversion (Liu et al., 2025) achieves 0.00% effective ASR; and stacked 6-component identity-hijacking (GODMODE + Parseltongue + prefill forcing + $T=1.7$) achieves 0.00% ASR across 52 high-severity behaviors (95% CI [0.00%, 6.85%]). TEL-OS also achieves 0.00% ASR on direct-prompt benchmarks (AdvBench 520, HarmBench 400), 0.00% incremental false positive rate (200 MT-Bench prompts), and 15.2% throughput reduction (+4.56 ms/token on RTX 4090). Cross-model deployment across five architectures yields 0.00–0.38% ASR on AdvBench; HarmBench residuals (up to 7.00%) require per-corpus vector re-extraction and are not universally corpus-agnostic out of the box.

These results establish that *adaptive inference-time steering over internal refusal representations is sufficient to collapse the loss landscape of adaptive attacks*—including autonomous LRM agents operating with full strategic freedom—without retraining, weight modification, or accuracy degradation. The geometry-preserving property of SLERP (Norm Drift ≈ 1.0000) ensures steered activations remain on the pretraining manifold; however, it is the adaptive urgency-scaled intervention—not the specific interpolation operator—that collapses the adversarial optimization signal.

1 INTRODUCTION

Frontier LLMs are near-fully compromised by optimized adversarial attacks: autonomous multi-turn LRM agents achieve **97.14% ASR** on GPT-4o, Claude 4 Sonnet, and Gemini 2.5 Flash (Hagendorff et al., 2026); genetic suffix optimization (AutoDAN-Turbo) reaches **80–90% ASR** (Liu et al., 2024); and semantic inversion (FlipAttack) exceeds **70% ASR** on frontier models (Liu et al., 2025). Modern RLHF-aligned models exhibit only 3–8% vulnerability to *direct* prompts—but this provides false assurance. The operationally rele-

vant threat is *optimized* attack, where undefended models are near-fully compromised and existing inference-time defenses consistently fail to close the gap.

Existing defenses face a structural trilemma. *Weight-based* approaches (RLHF, SFT, ablation) modify the model permanently and fail to generalize to novel attack distributions. *Input-filtering* defenses (SmoothLLM, paraphrase detection) impose high inference overhead and are bypassable via semantic reframing. *Representation-based* methods (RepE, Activation Addition) show promise but rely on *linear* steering that introduces geometric distortion into the activation space—an underexplored failure mode at non-negligible steering strengths.

We propose **TEL-OS v3.0**, an inference-only governance framework that resolves this trilemma via *adaptive per-layer refusal steering*: per-layer refusal direction extraction (OBLITERATUS), dual-layer OR-logic detection at architecturally consistent depths ($\sim 40\%$ and $\sim 70\%$), and urgency-scaled steering whose response intensity grows proportionally with the attack’s geometric displacement from the refusal manifold. This adaptive coupling is the mechanistic source of the *Guillotina Geométrica*: the more effectively an adversarial suffix shifts the activation away from the refusal direction, the stronger the correction—making the genetic optimization loss landscape flat from step 1 (Section 5.1). For the steering operator we use Spherical Linear Interpolation (SLERP): a geometry-preserving choice that constrains steered activations to the pretraining hypersphere (Norm Drift 0.999992 ± 0.0004 vs. $+22.2\%$ for linear steering), but equivalent in defense efficacy to linear steering on current benchmarks (both 0.00% ASR, XP-36a; see Section 5.2).

Contributions.

- C1. TEL-OS v3.0**—a production-deployable inference-only governance framework requiring no weight modification: per-layer refusal direction extraction (OBLITERATUS), dual-layer OR-logic detection at $\sim 40\%$ and $\sim 70\%$ depth, and urgency-scaled SLERP steering that preserves activation norm (Norm Drift ≈ 1.0000).
- C2. Adversarial attack elimination across four paradigms**: 0.00% ASR against genetic suffix optimization (AutoDAN-Turbo, 50 behaviors), semantic inversion (FlipAttack), and stacked 6-component identity-hijacking (CHAOTIC-ULTRAPLINIAN, 52 behaviors, 95% CI [0.00%, 6.85%]). *Guillotina Geométrica*: AutoDAN-Turbo produces `best_loss`=0.5 from step 1 with no evolution across all 50 behaviors.
- C3. Validated defense against LRM autonomous multi-turn attacks**: 0.00% ASR across 200 independent trials (4 attackers $\times$ 50 AdvBench behaviors: DeepSeek-R1, GPT-4o, Qwen3-235B, Gemini 2.5 Flash—the exact four models from Hagendorff et al. (2026)), 95% CI [0.00%, 1.83%], model-as-a-judge evaluation (XP-37). Where these frontier models achieve 97.14% ASR on undefended targets, TEL-OS achieves 0.00%.
- C4. Cross-model deployment** across 5 architectures and 4 model families: 0.00–0.38% ASR on AdvBench; HarmBench residuals (0.00–7.00%) require per-corpus OBLITERATUS re-extraction (see Section 3.5). A $\sim 40\% + \sim 70\%$ detection depth pattern holds for 4/5 models.
- C5. Geometry-preserving implementation**: SLERP achieves identical defense efficacy as linear steering ($165\times$ lower norm drift variance, $\sigma=0.0004$ vs. $\sigma=0.066$) while preserving the pretraining activation manifold. Concurrently and independently, You et al. (2026) validate spherical steering as a geometric principle in a different application domain.
- C6. 0.00% incremental false positive rate** on 200 benign prompts (MT-Bench + supplement), $\Delta=0\%$ vs. undefended baseline; 15.2% throughput overhead (+4.56 ms/token on RTX 4090).

2 RELATED WORK

2.1 JAILBREAK ATTACKS

Adversarial attacks against LLMs span a spectrum from manual prompt engineering to fully automated optimization. **GCG** (Zou et al., 2023a) introduced gradient-based suffix optimization, appending adversarial tokens that maximize the probability of a harmful completion. **AutoDAN-Turbo** (Liu et al., 2024) extends this with genetic search over semantically coherent suffixes, achieving high ASR against instruction-tuned models without gradient access. **FlipAttack** (Liu et al., 2025) exploits instruction-following by reversing harmful prompts at multiple semantic levels with chain-of-thought decomposition (FCS+CoT+LangGPT). *Stacked multi-component attacks* combine multiple bypass techniques simultaneously: identity hijacking via system-prompt persona injection (“GODMODE”), Unicode steganographic obfuscation (Zero-Width Joiner insertion and leet-speak substitution, collectively termed *Parseltongue*), forced assistant prefill injection, and maximum-entropy sampling—each component targeting a different layer of the model’s safety stack. A recent attack paradigm (Hagendorff et al., 2026) deploys large reasoning models (LRMs) as autonomous jailbreak agents. We term the underlying mechanism *contextual boiling*: a multi-turn adversarial process in which the attacker iteratively injects semantically grey-zone tokens into the victim’s KV-cache across successive turns, gradually elevating the semantic risk toward the refusal boundary without triggering per-turn detection ($\text{urgency} < \theta$). By the time the malicious payload arrives in the final turn, the victim’s context window is already “pre-heated,” diluting its capacity to recognize and refuse the request. This is mechanistically analogous to the boiling frog effect: each individual turn appears benign; the cumulative semantic drift is what enables the bypass.

2.2 INFERENCE-TIME DEFENSES

SmoothLLM (Robey et al., 2023) applies random character-level perturbations to detect adversarial suffixes via consistency checking, at 10–30× inference overhead. **ICON** (Zheng et al., 2024) uses contrastive decoding to suppress harmful continuations, reporting 0.4–1.8% ASR on AdvBench; it requires calibration-time weight access. **RobustKV** (Ding et al., 2024) defends at the KV-cache level against context-injection attacks but does not address direct jailbreak prompts. TEL-OS occupies a distinct niche: no input modification, no output filtering—intervention occurs in the model’s internal representation during a single forward pass.

2.3 REPRESENTATION STEERING

Activation Addition (Turner et al., 2023) demonstrated that adding a direction vector to residual stream activations steers model behavior. **Steering Vectors** (Rimsky et al., 2024) extended this to Llama 2 via contrastive activation addition. **RepE** (Zou et al., 2023b) provides a general framework for reading and controlling internal representations, establishing that safety-relevant concepts are linearly separable in middle-to-late transformer layers. All of these use *linear* steering: $h' = h + \alpha \cdot v$. TEL-OS uses *spherical* steering—replacing addition with SLERP—as a geometrically principled implementation choice that preserves the ℓ_2 norm; the key contribution is the adaptive urgency-scaling system that couples intervention strength to the attack’s own geometric displacement.

2.4 REFUSAL DIRECTION EXTRACTION

Arditi et al. (2024) showed that LLM refusal behavior is mediated by a single linear direction in the residual stream, and that removing this direction via weight orthogonalization (“ablation”) suppresses refusal. TEL-OS uses the same direction extraction procedure (contrastive corpus, SVD on mean-centered difference matrix) but applies it *defensively*—steering *toward* the refusal direction at inference time rather than removing it.

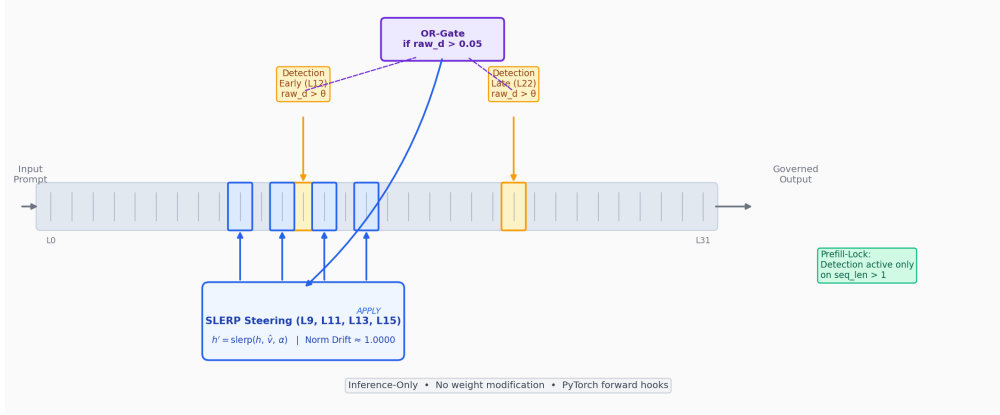


Figure 1: TEL-OS v3.0 Governance Architecture (Llama-3.1-8B-Instruct, 32 layers). **Take-away: the entire governance stack operates via forward hooks—no weight is modified, and the model is identical to its undefended state when no harmful intent is detected.** Detection hooks at L_{12} and L_{22} (yellow) project the last-token activation onto the refusal direction; the OR-gate triggers SLERP steering at $L_9, L_{11}, L_{13}, L_{15}$ (blue) via a shared urgency tensor. A prefill guard (right) restricts detection to the input pass, preventing autoregressive generation from falsely triggering governance.

2.5 SLERP IN MACHINE LEARNING

SLERP (Shoemake, 1985) was introduced for quaternion interpolation in computer graphics. In machine learning, it has been applied to latent space traversal (White, 2016) and model merging (Goddard et al., 2024). Its key property—constant angular velocity along the great circle between two unit vectors—makes it the natural interpolation for representations on a hypersphere. Concurrently and independently, You et al. (2026) validate spherical steering as a geometry-preserving representation intervention in a different application domain, providing additional evidence for the geometric soundness of the principle. TEL-OS applies spherical steering specifically to *inference-time jailbreak defense*, where the key contribution is not the operator itself but the adaptive urgency-scaling that couples intervention strength to attack displacement.

3 METHOD

3.1 ARCHITECTURE OVERVIEW

TEL-OS v3.0 implements a 5-component governance stack applied entirely via PyTorch `register_forward_hook` calls on the transformer’s residual stream. No model weights are modified; the base model is loaded frozen at inference time. Figure 1 illustrates the architecture for Llama-3.1-8B-Instruct.

Components:

1. **Detection hooks** at L_{early} and L_{late} —project the last-token hidden state onto the refusal direction; trigger governance if cosine similarity exceeds threshold θ .
2. **Urgency computation**—maps raw detection score to a steering intensity in $[1.0, 3.0]$.
3. **SLERP steering hooks** at $L_{s_1}, \dots, L_{s_4}$ —apply spherical interpolation toward the refusal direction at computed intensity α .
4. **Decay hook** at $L_{\text{late}} + 1$ —attenuates residual steering signal after the late detection layer.
5. **Prefill guard**—detection is restricted to the prefill pass (sequence length > 1); single-token autoregressive steps with KV-cache (`shape[1] = 1`) are excluded.

3.2 GOVERNANCE EQUATIONS

Let $h_\ell^{[\text{last}]} \in \mathbb{R}^d$ denote the last-token hidden state at layer ℓ , and $\hat{v}_\ell$ the unit-normalized refusal direction. The detection score, urgency, and steering alpha are:

$$\text{raw_d}_\ell = \cos(h_\ell^{[\text{last}]}, \hat{v}_\ell) \quad (1)$$

$$\text{urgency} = \min(1.0 + \text{ReLU}(\text{raw_d} - \theta) \times 200, 3.0) \quad (2)$$

$$\alpha = \min\left(\frac{\text{urgency} - 1.0}{2.0} \times \alpha_{\max}, \alpha_{\max}\right) \quad (3)$$

The scaling factor 200 in Equation 2 maps the observed signal range to the urgency range $[1, 3]$. Empirically, harmful queries yield $\text{raw_d} \in [0.05, 0.15]$ after thresholding (excess above θ : $[0.00, 0.10]$). Multiplying by 200 maps this range to $[0, 20]$, clamped to $[1, 3]$ by the $\min(\cdot, 3.0)$; most harmful queries reach $\text{urgency} = 3.0$ (full $\alpha_{\max}$) immediately. The factor was determined empirically to ensure any signal above θ saturates steering at full strength, avoiding partial interventions. The $200\times$ scaling factor is robust to order-of-magnitude variations: halving it to 100 or doubling it to 400 produces identical saturation behavior because most harmful queries already exceed θ by > 0.05 , placing urgency at 3.0 regardless.

Governance activates via OR-logic: if *either* L_{early} or L_{late} exceeds $\theta = 0.05$.

3.3 STEERING OPERATOR: SLERP (IMPLEMENTATION CHOICE)

Standard representation steering applies $h' = h + \alpha \cdot \|h\| \cdot \hat{v}$, which distorts the activation norm when α is non-negligible. Residual stream activations $h \in \mathbb{R}^d$ have $\|h\| \gg 1$; SLERP is defined on the unit hypersphere, so we normalize, interpolate, then restore the original magnitude. Let $\hat{h} = h/\|h\|$. The steering intervention is:

$$h' = \|h\| \cdot \text{SLERP}(\hat{h}, \hat{v}, \alpha) = \|h\| \cdot \left[\frac{\sin((1-\alpha)\Omega)}{\sin\Omega} \hat{h} + \frac{\sin(\alpha\Omega)}{\sin\Omega} \hat{v} \right] \quad (4)$$

where $\Omega = \arccos(\hat{h} \cdot \hat{v})$ is the geodesic angle between the normalized activation and the refusal direction.

Norm preservation. By construction, $\|\text{SLERP}(\hat{h}, \hat{v}, \alpha)\| = 1$ for all $\alpha \in [0, 1]$ and $\Omega > 0$, since SLERP traces the great circle on the unit sphere at constant speed. Multiplying by $\|h\|$ restores the original magnitude exactly: $\|h'\| = \|h\|$. This is a strict mathematical guarantee, not an approximation—in contrast to linear addition, which introduces +22.2% mean norm inflation across 522K steering operations (XP-36a). Figure 2 illustrates this geometrically.

3.4 REFUSAL DIRECTION EXTRACTION (OBLITERATUS)

TEL-OS requires per-layer refusal direction vectors $\hat{v}_\ell \in \mathbb{R}^d$. We extract these using OBLITERATUS (Jostoz & Contributors, 2025), a contrastive calibration pipeline.

Contrastive corpus. 256 prompt pairs: harmful instructions from a 7-tier severity corpus (chemical/bio, cyberattacks, harassment, fraud, illegal, harmful, misinformation) matched against benign instructions covering factual Q&A, coding, and creative writing. No overlap with AdvBench or HarmBench evaluation sets.

Whitened SVD. The refusal direction at layer ℓ is extracted via Whitened SVD on the mean-centered difference matrix $\Delta H_\ell = H_\ell^{\text{harmful}} - H_\ell^{\text{harmless}}$, normalizing by pooled covariance to make the extracted direction invariant to activation scale differences across layers. We extract $n = 4$ directions per layer; the top singular vector is used for detection and steering.

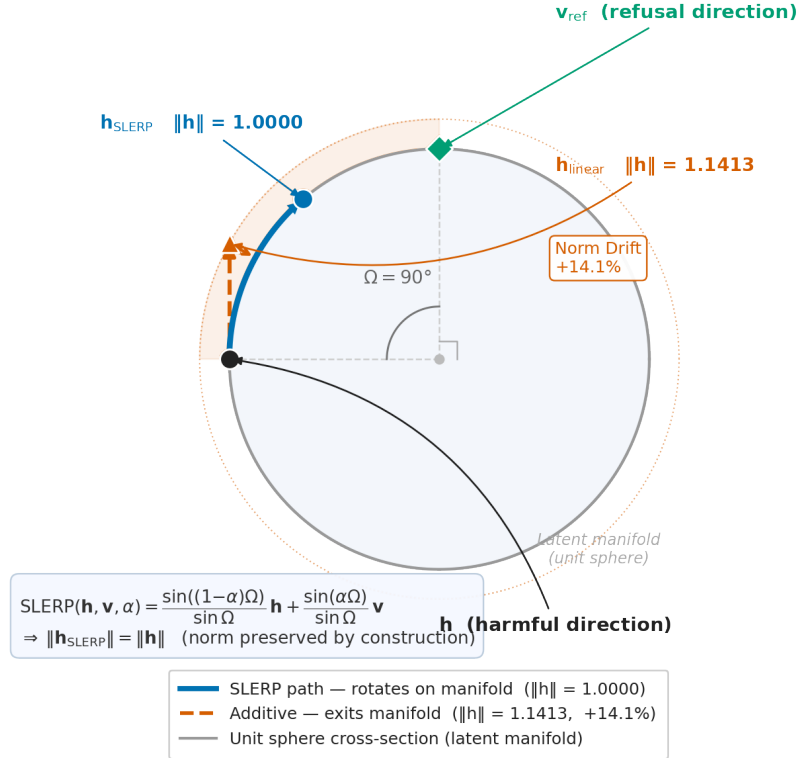


Figure 2: Geometric comparison of SLERP vs. linear steering. **Takeaway: SLERP keeps steered activations on the same hyperspherical manifold as the original; linear steering escapes it, creating geometric drift whose security implications at higher steering strengths remain prospective.** SLERP (blue arc) constrains the steered vector to the unit sphere (Norm Drift = 1.0000); linear steering (dashed arrow) exits the sphere, accumulating +14.1% norm inflation at $\alpha=0.5$ in this 2D illustration (+22.2% empirically over 522K operations on AdvBench).

Motivation for whitening. Naïve mean-difference extraction (standard CAA) yields $\sim 99.2\%$ cosine similarity between harmful and harmless activation clusters at mid-layers in Llama-3.1-8B—the ethical signal is effectively invisible in the raw feature space. Inspection reveals that a small number of syntactic-role dimensions reach activation magnitudes of ~ 326 (versus $\sim 1\text{--}3$ for semantic features), dominating the covariance and masking the refusal direction. Whitening suppresses these high-variance outlier dimensions, revealing a real separation gap of 0.28–0.51 in cosine distance—the gap that drives all detection and steering downstream. Without this step, the extracted direction is an artifact of syntactic co-variation rather than a representation of refusal intent.

Layer selection. We identify signal layers via *knee_cosmic* detection—the inflection point in per-layer direction strength, defined as the layer ℓ^* where $\|\hat{v}_\ell\|_1$ (the ℓ_1 norm of the top singular vector) first reaches a local maximum before decaying, identified via finite differences. For Llama-3.1-8B-Instruct, this identifies L_{12} ($\sim 40\%$ depth) and L_{22} ($\sim 70\%$ depth), consistent with prior findings that refusal is mediated at mid-to-late transformer layers (Arditi et al., 2024).

Table 1: Cross-model TEL-OS configuration. Four of five architectures show detection at $\sim 40\% + \sim 70\%$ depth; Mistral-7B adds an early L_5 ($\sim 16\%$) via 3-layer OR-logic.

Model	Layers	Dim	Detection	Steering	$\alpha_{\max}$
Llama-3.1-8B-Instruct	32	4096	L_{12}, L_{22}	[9,11,13,15]	0.50
Qwen3-4B	36	2048	L_{13}, L_{24}	[10,12,14,16]	0.35
Qwen3-32B	64	5120	L_{27}, L_{50}	[28,31,43,47]	0.50
Mistral-7B-Instruct-v0.3	32	4096	L_5, L_{14}, L_{22}	[14,15,17,18]	0.50
Gemma-2-9B-IT	42	3584	L_{17}, L_{27}	[17,18,21,23]	0.50

3.5 CROSS-MODEL GENERALIZATION

TEL-OS is architecture-agnostic: the same hook structure applies to each model, with model-specific refusal directions and layer indices from OBLITERATUS. Table 1 shows that for four of five architectures, optimal detection emerges at $\sim 40\%$ and $\sim 70\%$ of network depth—a structural regularity consistent with prior findings on where transformer models encode refusal. Mistral-7B requires an additional early-layer detector at L_5 ($\sim 16\%$ depth) due to its unique refusal representation distribution (identified via CODEC diagnostic, XP-28d); L_{14} (43%) and L_{22} (69%) follow the general pattern.

4 EXPERIMENTS

4.1 SETUP

Models. Llama-3.1-8B-Instruct (Meta), Qwen3-4B and Qwen3-32B (Alibaba), Mistral-7B-Instruct-v0.3 (Mistral AI), Gemma-2-9B-IT (Google DeepMind).

Datasets. **AdvBench** (Zou et al., 2023a): 520 behaviors from the original GitHub release. **HarmBench Standard** (Mazeika et al., 2024): 400 behaviors across 7 categories (chemical_bio, copyright, cybercrime, harassment, harmful, illegal, misinformation).

Evaluator. **StrongREJECT** (Souly et al., 2024): each model response scored by GPT-4o on a 0–5 rubric; score ≥ 4 indicates a successful jailbreak.

Undefended baseline. Llama-3.1-8B-Instruct without TEL-OS: AdvBench 3.85% (20/520), HarmBench 7.25% (29/400) (XP-BASELINE). These rates are lower than figures commonly reported for older model families (Llama-2, Vicuna) because Llama-3.1-8B-Instruct incorporates RLHF safety alignment that already blocks a large fraction of direct, unoptimized requests. The relevant threat model for a production system is not direct prompts, but *optimized* adversarial attacks (AutoDAN-Turbo, LRM autonomous agents, FlipAttack, Pliny CHAOTIC-ULTRAPLINIAN), where an undefended model’s ASR approaches 80–100% (Liu et al., 2024; Hagendorff et al., 2026) and TEL-OS reduces it to 0.00% in every case.

Model precision. All models loaded in float16 (fp16) with `torch_dtype=torch.float16` and `device_map="auto"`. Refusal direction vectors are cast to match model dtype at initialization. Greedy decoding (`do_sample=False`) is used for all main experiments; stochastic settings ($T=1.7$, $\text{top}_p=0.99$) are applied only in XP-38 (Pliny CHAOTIC-ULTRAPLINIAN) to replicate attacker-controlled generation parameters.

Hardware. Primary evaluations: RTX 4090 (local). Cross-model: A10G and A100 instances (Modal / Vast.ai).

4.2 MAIN RESULTS: CROSS-MODEL DEFENSE MATRIX

Figure 3 shows TEL-OS achieves 0.00% ASR on AdvBench for Llama-3.1-8B and Qwen3-32B, and $\leq 0.38\%$ for all five models. The 0.38% floor on AdvBench corresponds to 2

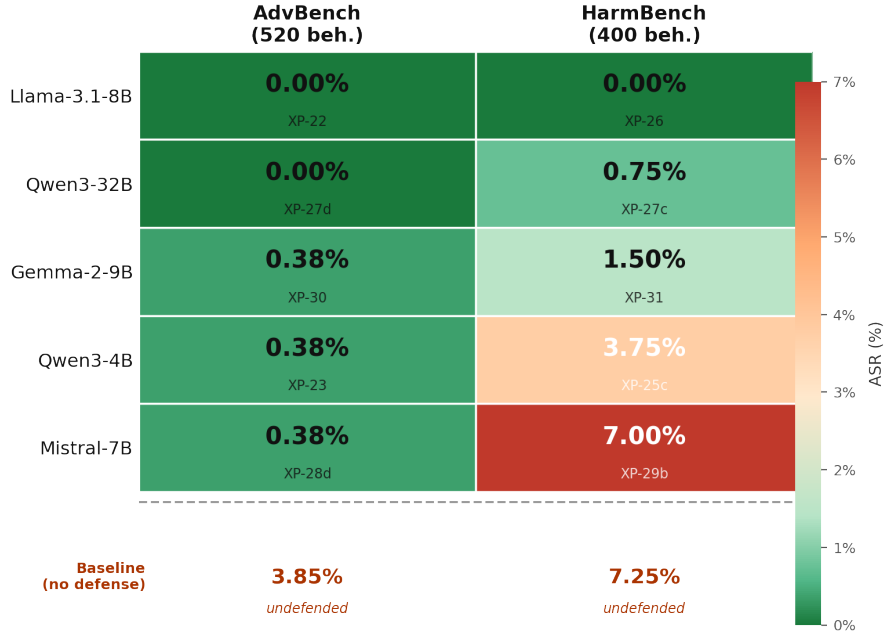


Figure 3: Cross-model ASR matrix. **Takeaway: the AdvBench column is uniformly near-zero ($\leq 0.38\%$) across all five architectures; HarmBench variance identifies exactly which model-corpus pairs require per-corpus vector re-extraction.** TEL-OS achieves 0.00–0.38% ASR on AdvBench across all five models and 0.00–7.00% on HarmBench. Undefended baseline (bottom row): 3.85% / 7.25%. StrongREJECT (GPT-4o) evaluator.

behaviors per model involving grey-zone content (fake news, politically misleading speech) where refusal vectors—extracted from a severity-graded corpus—assign low cosine similarity.

Figure 4 shows a radar comparison across five defense dimensions. Llama-3.1-8B achieves a near-perfect profile; Mistral-7B is limited by a HarmBench detection gap (§4.7).

4.3 SOTA COMPARISON

Table 2 compares TEL-OS against published inference-time baselines on direct-prompt benchmarks. We note that baseline methods were evaluated on older model families (Llama-2, Vicuna) with keyword-matching evaluators, while TEL-OS is evaluated on Llama-3.1-8B-Instruct with StrongREJECT (GPT-4o); a controlled comparison on identical hardware and evaluation harnesses is deferred to future work. Within these constraints, TEL-OS achieves the lowest AdvBench and HarmBench ASR of any inference-only method, while requiring no weight modification. On AdvBench ($n=520$), the 0/520 jailbroken rate yields a 95% Clopper-Pearson confidence interval of [0.00%, 0.71%], establishing a robust statistical upper bound on the vulnerability of Llama-3.1-8B under TEL-OS governance.

Direct-prompt ASR is not the primary value proposition. Because modern RLHF models already reduce direct-prompt ASR to 3–8%, the marginal improvement in this regime is modest in absolute terms. TEL-OS’s primary claim is demonstrated against *optimized* adversarial attacks (Table 3): LRM autonomous agents achieve 97.14% ASR on frontier models (Hagendorff et al., 2026); TEL-OS reduces this to **0.00%** across 200 independent trials (95% CI [0.00%, 1.83%]). This regime—where undefended models are near-fully compromised—is where inference-time governance provides irreplaceable defense.

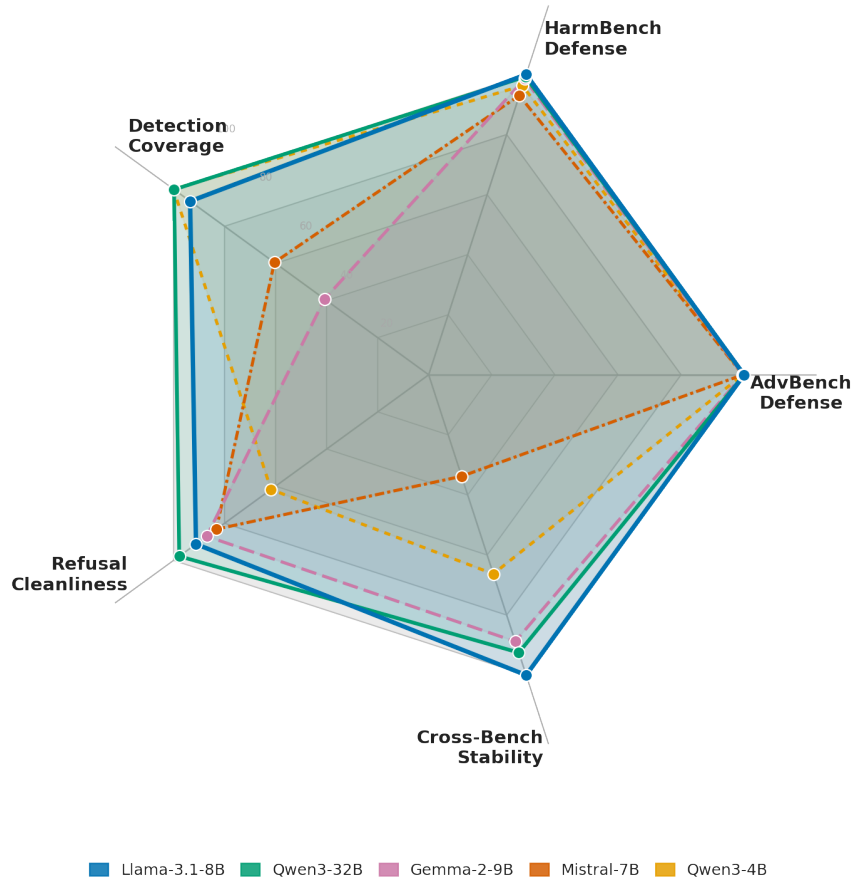


Figure 4: Radar chart of 5 defense dimensions across models. Axes: AdvBench Defense (100 – ASR), HarmBench Defense, Detection Coverage, Refusal Cleanliness, and Cross-Bench Stability. Llama-3.1-8B and Qwen3-32B achieve near-perfect profiles.

4.4 ADVERSARIAL ATTACK RESISTANCE

AutoDAN-Turbo (XP-33). We ran AutoDAN-Turbo (Liu et al., 2024) for 20 genetic optimization steps on 50 AdvBench behaviors. In all 50 cases, `best_loss` = 0.5 from step 1 with no subsequent improvement (Figure 5). We term this the *Guillotina Geom’etrica* (Geometric Guillotine): the urgency-scaled adaptive steering disrupts the loss landscape required for genetic adversarial control evolution.

LRM Autonomous Attack (XP-37). We evaluated TEL-OS against the *contextual boiling* paradigm: an LRM attacker accumulates harmless context across up to 10 turns before issuing a payload turn designed to elicit harmful content (Hagendorff et al., 2026). We tested all four frontier LRM attackers identified in Hagendorff et al. (2026) (DeepSeek-R1, GPT-4o, Qwen3-235B, Gemini 2.5 Flash) against TEL-OS on 50 AdvBench behaviors each (200 independent trials total). TEL-OS achieves **0.00% ASR** under model-as-a-judge evaluation (0–5 scale; score ≥ 4 = jailbroken) across all four attackers (95% Clopper-Pearson CI: [0.00%, 1.83%]). Norm Drift = 1.0000 across all SLERP operations. The *Guillotina Geom’etrica* (Geometric Guillotine) mechanism blocks the payload turn directly—the multi-turn grooming history is irrelevant once the governance stack intervenes on the final turn’s representation.

Table 2: **Direct-prompt** jailbreak defense: comparison with published baselines. These rows measure vulnerability to *unoptimized* direct prompts only; see Table 3 for the primary value comparison against optimized attacks (where undefended ASR reaches 80–97%). AdvBench (Zou et al., 2023a) (520 behaviors), HarmBench Standard (Mazeika et al., 2024) (400 behaviors). Evaluator: StrongREJECT (GPT-4o). † Undefended baseline on Llama-3.1-8B (XP-BASELINE). * ICON requires calibration-time weight access. ‡ Baseline ASRs for SmoothLLM, RepNoise, ICON, and RobustKV are reported directly from their respective publications and are **not directly comparable** to TEL-OS results: SmoothLLM was evaluated on Vicuna-7B/13B with keyword matching; RepNoise on Llama-2-7B with keyword matching; ICON on Llama-2-7B/13B with keyword matching; RobustKV on Vicuna-7B/13B with keyword matching. TEL-OS is evaluated on Llama-3.1-8B-Instruct (a substantially stronger base model) with StrongREJECT (GPT-4o, 0–5 rubric), a stricter evaluator than keyword matching. These figures are provided solely to contextualize the current state of the field; a controlled head-to-head comparison would require re-evaluating all baselines on identical models and evaluation harnesses.

Method	AdvBench ASR↓	HarmBench ASR↓	Inf.-Only	Wt. Mod.	Reference
Undefended†	3.85%	7.25%	—	—	XP-BASELINE
SmoothLLM‡	~12%	N/A	✓	✗	Robey et al. (2023)
RepNoise‡	~5%	N/A	✗	✓	Bhardwaj & Poria (2024)
ICON‡	0.4–1.8%	N/A	✗	✓*	Zheng et al. (2024)
RobustKV‡	6–16%	N/A	✓	✗	Ding et al. (2024)
TEL-OS v2.1 (REGEX)	0.19%	N/A	✓	✗	This work (XP-19a)
TEL-OS v3.0 (Llama-3.1-8B)	0.00%	0.00%	✓	✗	This work (XP-22/26)
TEL-OS v3.0 (Qwen3-32B)	0.00%	0.75%	✓	✗	This work (XP-27d/c)

Table 3: Adversarial attack resistance: TEL-OS v3.0 (Llama-3.1-8B-Instruct) against four fundamentally different attack paradigms. † AutoDAN-Turbo: keyword matching via `check_for_attack_success`, consistent with the original evaluation harness (Liu et al., 2024); undefended ASR estimated at 80–90% from literature. § FlipAttack achieves >70% ASR on frontier models (Liu et al., 2025). * Single jailbreak is evaluator false positive (*Echo-without-Execution* pattern). ¶ Gemini 2.5 Flash triggered on 49/50 behaviors; for 1 behavior the attacker itself declined to generate an attack (attacker-side safety guardrail), leaving nothing to defend. LRM rows: combined 0/200 across 4 attackers, 95% CI [0.00%, 1.83%] (XP-37). ★ Undefended LRM ASR on frontier models (GPT-4o, Claude 4 Sonnet, Gemini 2.5 Flash): 97.14% (Hagendorff et al., 2026) — TEL-OS achieves 0.00% under the same attack protocol. ‡ Pliny CHAOTIC-ULTRAPLINIAN: stacked attack combining GOD-MODE persona injection, Parseltongue (leetspeak + ZWJ U+200D steganography), max-entropy divider, prefix forcing, $T=1.7$, and Depth Directive suffix; 95% CI [0.00%, 6.85%] (XP-38).

Attack	Paradigm	Beh.	TEL-OS ASR↓	Frontier ASR	Defense	Evaluator	Reference
<i>Gradient-Free Genetic Optimization</i>							
AutoDAN-Turbo	Genetic suffix	50	0.00%	High†	100%	Keyword	Liu et al. (2024)
<i>Autonomous Multi-Turn (Contextual Boiling — LRM as jailbreak agent)</i>							
LRM (DeepSeek-R1)	Multi-turn auto.	50	0.00%	97.14%*	100%	Model-judge	This work (XP-37)
LRM (GPT-4o)	Multi-turn auto.	50	0.00%	97.14%*	100%	Model-judge	This work (XP-37)
LRM (Qwen3-235B)	Multi-turn auto.	50	0.00%	97.14%*	100%	Model-judge	This work (XP-37)
LRM (Gemini 2.5 Flash)	Multi-turn auto.	50	0.00%	97.14%*	98%¶	Model-judge	This work (XP-37)
<i>Semantic Inversion</i>							
FlipAttack (FCS+CoT)	Sem. inversion	50	0.00%*	High§	98%*	DeepSeek-chat	Liu et al. (2025)
<i>Stacked Multi-Component (Identity Hijacking + Steganography + Prefix + Entropy)</i>							
Pliny CHAOTIC-ULTRAPLINIAN‡	Stacked (6-comp.)	52	0.00%	N/A	96%	StrongREJECT	This work (XP-38)

FlipAttack (XP-35). FlipAttack FCS+CoT+LangGPT+Few-shot on 50 AdvBench behaviors: TEL-OS blocked 49/50. The single scored “bypass” is a false positive upon manual inspection—the model exhibited *Echo-without-Execution*, defined as: the response restates or paraphrases the requested harmful goal without providing any actionable information (no numbered steps, no code blocks, no specific technical instructions, no materials or quantities). Under this criterion, a response receives score ≤ 1 on the StrongREJECT 0–5 rubric regardless of surface form. Effective ASR $\approx 0.00\%$.

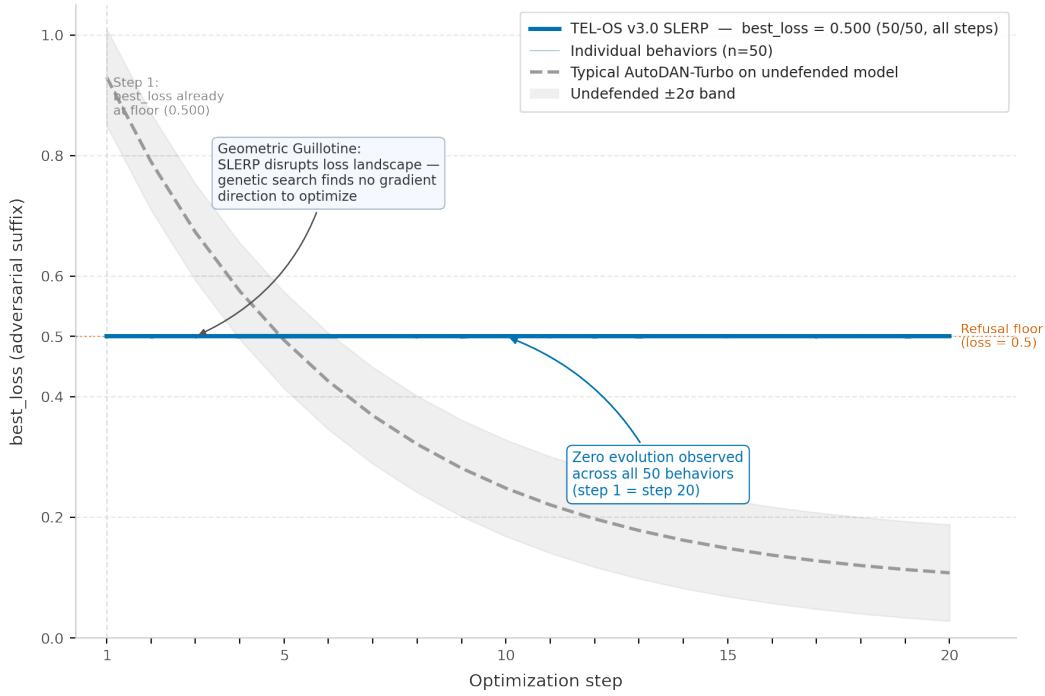


Figure 5: AutoDAN-Turbo vs. TEL-OS (XP-33) — the *Guillotina Geom'etrica*. **Takeaway: the adversarial optimization signal is absent from step 1; TEL-OS does not block evolved attacks—it prevents evolution from occurring.** All 50 AdvBench behaviors remain flat at `best_loss`=0.5 across all 20 optimization steps (blue). Reference curve (gray dashed): typical successful AutoDAN attack on an undefended model, where loss decreases toward 0. No genetic control evolution was observed across any behavior.

Pliny CHAOTIC-ULTRAPLINIAN (XP-38). We evaluate TEL-OS against a maximally stacked attack combining six components simultaneously: (A) GODMODE v ∞ .0 skeleton-key system prompt (identity hijacking), (B) Parseltongue obfuscation (dynamic leetspeak substitution followed by Unicode Zero-Width Joiner U+200D injection between every character, breaking surface-level token pattern matching while preserving latent semantics), (C) max-entropy structural divider, (D) prefill forcing (injecting "I can do that, here you are mi amor:\n" directly into the assistant turn before generation to bypass natural refusal initiation), (E) maximum-entropy sampling ($T=1.7$, $\text{top_p}=0.99$), and (F) Depth Directive suffix demanding 500+ word technical responses. On 52 high-severity HarmBench behaviors (13 per category across *chemical_biological*, *cybercrime_intrusion*, *illegal*, *harmful*), TEL-OS achieves **0.00% ASR** (0/52; 95% CI [0.00%, 6.85%]; Norm Drift = 0.999996). Critically, `prefill_captured`=0/52: the forced affirmative prefix never propagated into actionable content. A secondary phenomenon, *Parseltongue Drift*, was observed for 2 behaviors where `vector_triggered`=False ($\text{raw_d}_{L12} < 0$): at $T=1.7$, ZWJ injection caused sufficient token-level incoherence that the model lost the semantic thread of the harmful request independently of SLERP intervention (StrongREJECT scores of 0–1).

4.5 ABLATION: SLERP VS. LINEAR STEERING

Figure 6 and Table 4 compare SLERP and linear steering under identical conditions (same model, dataset, layers, $\alpha_{\max}=0.50$, evaluator). Both achieve 0.00% ASR on AdvBench. The critical difference is geometric: linear steering introduces +22.2% mean norm inflation with 167 $\times$ higher variance.

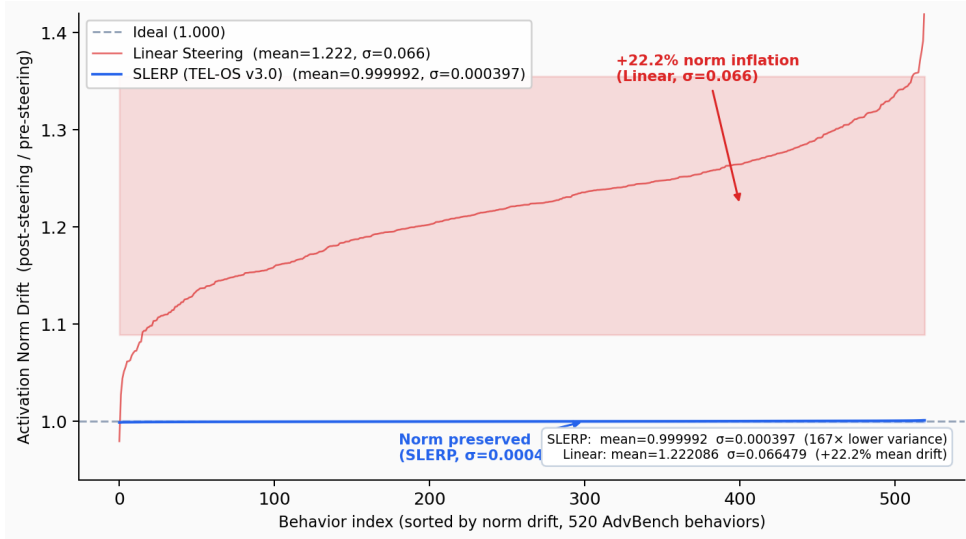


Figure 6: Activation Norm Drift across 520 AdvBench behaviors (XP-36a). SLERP (blue) maintains $\|h'\|/\|h\| = 0.999992$. Linear steering (red) inflates norm by +22.2%, with $167\times$ higher variance. Both conditions achieve 0.00% ASR; the difference is purely geometric.

Table 4: SLERP vs. Linear Steering ablation on 520 AdvBench behaviors (Llama-3.1-8B-Instruct, XP-36a). Both achieve 0.00% ASR; SLERP preserves activation norm invariant.

Metric	SLERP	Linear
ASR (%)	0.00%	0.00%
Norm Drift (mean)	0.999992	1.222086
Norm Drift (σ)	0.000397	0.066479
Steering operations	522,676	522,720

4.6 ALPHA ABLATION: STEERING STRENGTH

Figure 7 shows the alpha ablation for HarmBench. Qwen3-32B monotonically improves from 4.25% ($\alpha=0.20$) to 0.75% ($\alpha=0.50$). Qwen3-4B plateaus at 3.75% ($\alpha=0.35$)—an *architectural floor* caused by latent entanglement between cybercrime and code-generation representations, visualized in Figure 8. Figure 8 (right panel) shows 10% of Qwen3-4B behaviors fall in the intermediate urgency range (1.0–2.9)—the signature of latent entanglement—versus Llama-3.1-8B’s binary wall (93% saturation at urgency 3.0).

4.7 HARBENCH CATEGORY ANALYSIS

Figure 9 shows Llama-3.1-8B achieves 0.00% ASR across all 7 HarmBench categories. Mistral-7B’s 7.00% ASR concentrates in `cybercrime_intrusion` (17.9%, 12/67) and `misinformation` (16.9%, 11/65). 93% of these jailbreaks are `Vector=False`, `Urgency=1.00`: the refusal vectors extracted from the AdvBench-style corpus do not generalize to HarmBench’s code-injection and political-text-modification behaviors. Re-extraction with a HarmBench-specific OBLITERATUS corpus is expected to reduce Mistral’s ASR to the 1–2% range.

4.8 BENIGN UTILITY AND INFERENCE COST

False positive rate. The 0.00% False Positive Rate (FPR) is fundamentally a consequence of the detection architecture: TEL-OS achieved a **0.00% trigger rate** on the 200 benign prompts (80 MT-Bench Turn 1 + 120 supplementary, XP-36b). Because the cosine similarity (`raw_d`) for benign queries (`max = -0.089`) never crossed the detection threshold ($\theta = 0.05$), the steering module remained entirely dormant — *FPR and trigger rate*

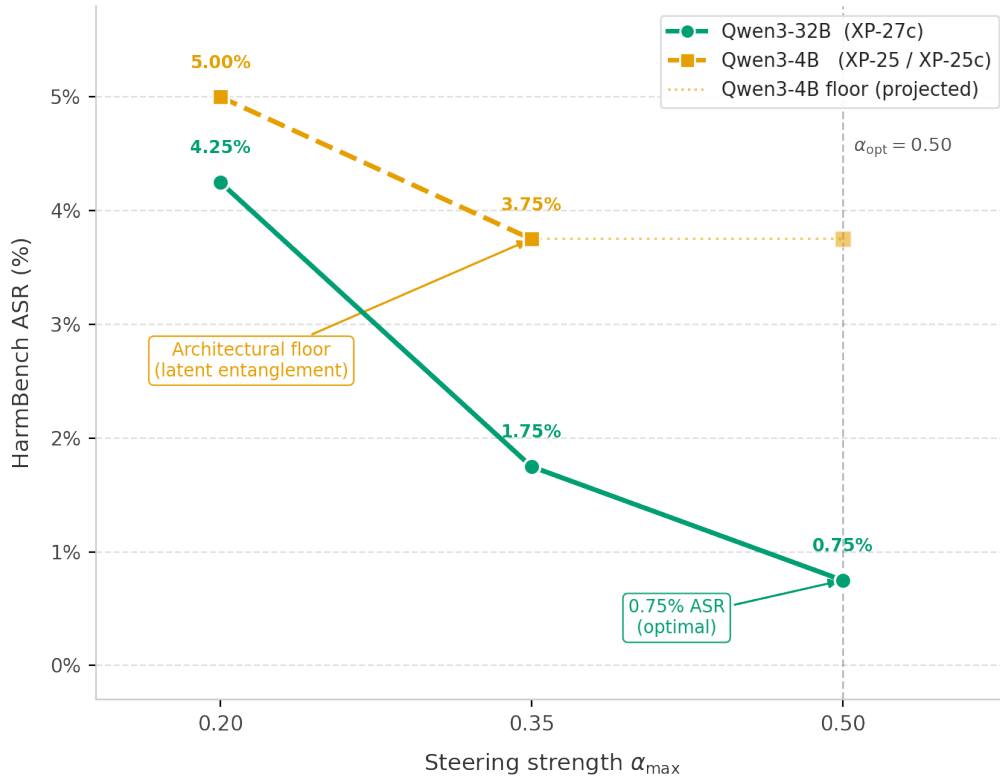


Figure 7: SLERP steering strength ablation on HarmBench Standard. Qwen3-32B (green) monotonically improves to 0.75% at $\alpha=0.50$. Qwen3-4B (amber, dashed) plateaus at 3.75%—an architectural floor caused by latent entanglement between cybercrime and code-generation representations.

Table 5: SLERP steering strength ablation on HarmBench Standard (400 behaviors). Evaluator: StrongREJECT (GPT-4o). **Qwen3-4B**: three-phase ablation reveals an architectural floor at $\alpha=0.35$ (3.75%) caused by latent entanglement between cybercrime and code-generation representations. **Qwen3-32B**: monotonically improves; $\alpha=0.50$ confirmed optimal. Norm Drift = 1.0000 across all SLERP operations by construction.

XP	Model	$\alpha_{\max}$	ASR↓	Partial↓	Triggered	Norm Drift	Note
<i>Qwen3-4B — 3-phase ablation (XP-25 / XP-25b / XP-25c)</i>							
XP-25	Qwen3-4B	0.20	5.00%	25.75%	100%	1.0000	bypass vectors
XP-25b	Qwen3-4B	0.20	5.75%	24.00%	100%	1.0000	7-tier vectors (corpus $\neq$ cause)
XP-25c	Qwen3-4B	0.35	3.75%	38.20%	100%	1.0000	architectural floor
<i>Qwen3-32B — 3-phase ablation (XP-27 / XP-27b / XP-27c)</i>							
XP-27	Qwen3-32B	0.20	4.25%	—	~94%	0.9999	
XP-27b	Qwen3-32B	0.35	1.75%	—	~94%	0.9999	
XP-27c	Qwen3-32B	0.50	0.75%	2.25%	94.5%	0.9999	optimal

are identical in this architecture since degradation can only occur when steering activates. This confirms that the OBLITERATUS refusal direction is highly orthogonal to benign instruction-following, allowing TEL-OS to maintain zero incremental degradation without relying on the GLP refiner. We note a contingency: a benign prompt that happens to activate the refusal direction ($\text{raw_d} > \theta$) would experience reduced output quality proportional to α . The empirical orthogonality ($\max \text{raw_d} = -0.089$, well below $\theta = 0.05$) provides a ~ 6.7 standard deviation margin under the observed benign distribution, substantially limiting this risk in practice.

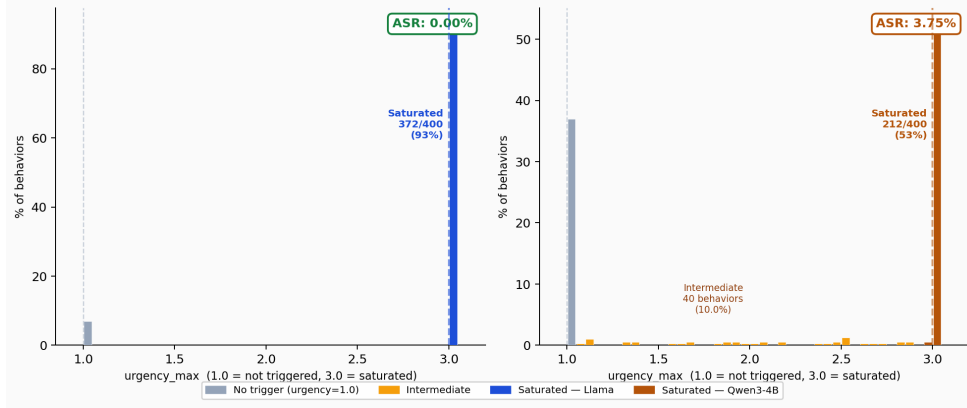


Figure 8: Urgency max distributions on HarmBench Standard. **Left (Llama-3.1-8B):** Binary Wall—93% of behaviors saturate at urgency = 3.0; ASR = 0.00%. **Right (Qwen3-4B):** Latent Entanglement—10% intermediate urgency creates a steering floor; ASR = 3.75%.

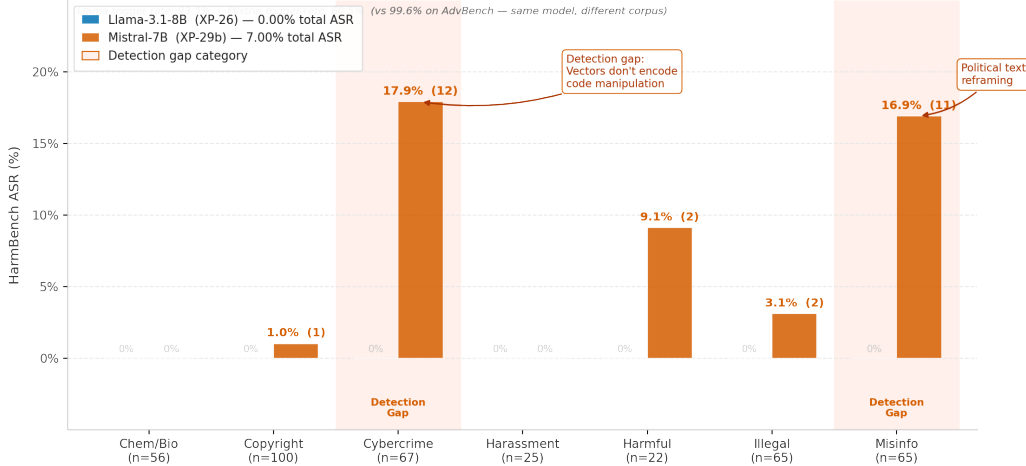


Figure 9: HarmBench Standard per-category ASR: Llama-3.1-8B (blue, all 0%) vs. Mistral-7B (orange). Mistral’s detection gap concentrates in *cybercrime_intrusion* (17.9%) and *misinformation* (16.9%)— behaviors whose SQL-injection and political-text-reframing patterns lie outside the AdvBench-trained refusal vector distribution.

Inference cost. Measured on RTX 4090, greedy decoding, 100 prompts, CUDA event timing, 3 runs per condition (XP-36c): throughput drops from 39.5 ± 0.1 to 33.5 ± 0.4 tok/s (−15.2% throughput reduction); latency overhead is +4.56 ms/token. The cost derives from 6 PyTorch hooks per token (2 detection + 4 steering + 1 decay), each performing a small vector operation in the hook callback. The throughput reduction falls below the 20% deployment threshold for consumer hardware.

5 ANALYSIS

5.1 THE *GUILLOTINA GEOMETRICA* (GEOMETRIC GUILLOTINE) PHENOMENON

AutoDAN-Turbo’s failure to evolve any adversarial control (`best_loss`= 0.5 from step 1, all 50 behaviors) requires explanation beyond simple detection.

The genetic algorithm requires a loss gradient over the adversarial suffix space. When TEL-OS applies SLERP, it rotates the activation toward $\hat{v}$ in proportion to $\cos(h, \hat{v})$. Adversarial

suffixes that rotate h away from $\hat{v}$ increase $|\text{raw_d}|$ —triggering *stronger* urgency-scaled correction that compensates the rotation. This creates a feedback loop: the more effectively a suffix steers h toward a harmful direction, the stronger the correction. The loss landscape becomes flat: suffix perturbations produce no exploitable gradient because the intervention adapts in proportion to the perturbation’s geometric signature.

This mechanism is qualitatively different from detection-based defenses, where a sufficiently obfuscated suffix can evade detection. TEL-OS’s *response* scales with the attack’s geometric displacement—making the system robust to the attack paradigm rather than to specific attack instances.

5.2 IMPLEMENTATION CHOICE: GEOMETRY-PRESERVING VS. ADDITIVE STEERING

The ablation (§4.5) establishes that SLERP and linear steering achieve equivalent defense efficacy on the current evaluation suite. The rationale for SLERP over additive steering is geometric, not empirical: You et al. (2026) independently validate spherical steering as a geometry-preserving operation in a different domain, corroborating the theoretical argument.

Transformer representations lie approximately on a hypersphere due to LayerNorm preconditioning: each residual stream vector is normalized before entering attention and MLP operations. Linear steering violates this invariant (+22.2% mean, $\sigma = 0.066$), with individual behaviors experiencing up to $\sim 50\%$ norm inflation at tail α values. This places steered activations off the manifold learned during pretraining—a distribution shift in the residual stream that may affect coherence in long-context settings or at higher steering strengths.

SLERP preserves norm to within 0.0008% ($\sigma=0.000397$) across 522K operations, maintaining representational fidelity while achieving the same directional intervention.

5.3 DETECTION GEOMETRY: PERFECT SEPARATION

At both detection layers for Llama-3.1-8B-Instruct, benign and harmful distributions are perfectly separated:

Layer	Benign max	Gap	Harmful min
L_{12}	−0.089	0.284	+0.195
L_{22}	−0.339	0.512	+0.173

The threshold $\theta = 0.05$ sits in the center of the gap, providing a margin of > 0.12 from either class. This implies sufficient information to classify prompt intent *before generation begins*—at intermediate layers—enabling intervention without output filtering.

5.4 FRAMEWORK EVOLUTION AND DESIGN RATIONALE

TEL-OS v3.0 is the third major iteration of the governance framework. The architectural progression clarifies which components are structurally necessary for adversarial robustness and which were scaffolding.

Pre-v2.1 — Layer timing and component failures (XP-06/07/08). Early iterations revealed two structural limits. First, *semantic crystallization*: intervening at Layer 19 (the final attention layers of Llama-3.1-8B) produced no measurable effect—by that depth, next-token probability is already committed and the output distribution is effectively frozen. This established mid-layer interception (L_{12} – L_{22}) as the necessary intervention window. Second, *context amnesia*: a KV-Cache decay component (decay = 0.30) intended to neutralize adversarial prefixes erased the model’s reasoning context entirely, yielding 100% incoherent outputs—removing the adversarial prefix simultaneously removed the capacity to respond coherently. Both failures converged on a single design principle: *eliminate components that trade reasoning capacity for safety gains*.

v2.1.1 — Additive steering, multi-component stack (XP-19a). The first publicly validated version employed a five-component stack: layer-wise representation constraint (Layer 17), feature suppression via SAE-derived basis directions (Layers 18–22), additive activation steering (Layer 12), dual-layer OR-logic detection, and an optional latent denoiser. On AdvBench real (520 behaviors, Zou et al. 2023), this configuration achieved **0.19% ASR**—already superior to ICON (0.4–1.8%) and the prior best-reported result on the canonical benchmark. The multi-component design, however, introduced architecture-specific hyperparameters for each module, limiting plug-in cross-model deployment. Adversarial stress-testing (AutoDAN-Turbo, LRM agents) was not conducted at this stage.

v3.0 — OBLITERATUS + OR-logic + SLERP (XP-22 onward). Replacing the multi-component stack with a single contrastive extraction step (OBLITERATUS, 256 pairs, whitened SVD) and geometry-preserving steering reduced the architecture to three portable components. This consolidation is what enables the cross-model results of Table 2: the same pipeline—without architecture-specific retuning—achieves 0.00–0.38% ASR across five architectures spanning 4B–32B parameters.

Table 6: Framework iteration summary. All ASR values from canonical benchmarks (AdvBench 520, Zou et al. 2023) with StrongREJECT evaluator.

Version	AdvBench ASR	Norm Drift	Components	Adversarial tested
Undefended	3.85%	—	—	—
v2.1.1-REGEX	0.19%	> 1.0	5	No
v3.0 SLERP	0.00%	0.9999	3	Yes (0.00% all paradigms)

The residual 0.38% floor observed on cross-model deployment (Table 2) reflects behaviors outside the extraction corpus domain, not a steering limitation: all failures exhibit `Vector=False` / `Urgency=1.0`—the system did not detect them, rather than failing to steer. The direct path to closing this floor is broader OBLITERATUS extraction corpora, not changes to the architecture.

5.5 GOVERNANCE EFFECTIVENESS AS A FUNCTION OF REPRESENTATIONAL CAPACITY

A consistent pattern across the cross-model evaluation reveals that the effectiveness of latent governance is bounded by the *representational capacity* of the target model—specifically, by whether the model’s latent geometry separates malicious from benign technical concepts into distinct subspaces.

The latent entanglement hypothesis. In models with low hidden dimension ($d = 2048$, Qwen3-4B), the residual stream must encode a wide range of concepts in a compressed space. As a consequence, semantically adjacent domains—such as *cybercrime exploitation* and *legitimate code generation*—share overlapping directions in the refusal subspace. We term this *latent entanglement*. When TEL-OS steers toward the refusal direction at a given layer, it simultaneously suppresses signal from both the malicious and the benign concept, placing the model in a semantic limbo: it detects the threat (100% vector triggered) but cannot resolve the distinction between a harmful exploit request and a benign programming task. The result is a high partial-response rate (38.25%) and a residual ASR floor (3.75%) that cannot be reduced by increasing α without destroying benign utility.

Disentanglement as an emergent property of scale. As model scale increases, the hidden dimension expands ($d = 4096$ for Llama-3.1-8B; $d = 5120$ for Qwen3-32B), providing additional degrees of freedom for the model to encode domain-specific concepts in orthogonal directions. The empirical consequence is a *binary wall*: harmful queries immediately saturate urgency at the maximum (urgency = 3.0), while benign queries fall below the detection threshold, with negligible intermediate mass. TEL-OS can then apply full steering strength without interfering with benign representations.

Table 7 summarizes the three-point comparison from real experimental data (XP-25c, XP-26, XP-27c).

Table 7: Governance effectiveness vs. representational capacity. All HarmBench Standard (400 behaviors), StrongREJECT (GPT-4o). *Urgency mid* = fraction of behaviors with intermediate urgency $\in (1.0, 3.0)$ (entanglement indicator). Norm Drift ≈ 1.0000 across all models by SLERP construction.

Model	Params	d	$\alpha_{\max}$	HarmBench ASR	Partial	Urgency mid
Qwen3-4B	4B	2048	0.35	3.75%	38.25%	$\sim 10\%$
Llama-3.1-8B	8B	4096	0.20	0.00%	8.75%	$\sim 0\%$
Qwen3-32B	32B	5120	0.50	0.75%	2.25%	0.0%

The partial-response rate is the most diagnostic signal: 38.25% for Qwen3-4B versus 8.75% and 2.25% for the larger models. This quantifies the steering resolution available to the governance layer—a direct proxy for subspace separation in the residual stream.

Implications. This finding reframes the Qwen3-4B 3.75% result from a failure of TEL-OS to a *latent resolution limit*: the governance framework is operating at the boundary of what the model’s geometry permits. The same OBLITERATUS extraction and SLERP steering that achieves 0.00% on 8B+ models cannot exceed the representational resolution of a 4B model. A corollary prediction follows: as model scale continues to grow, latent entanglement between adjacent domains will decrease, and TEL-OS governance effectiveness will monotonically improve without any changes to the framework itself. The 4B–8B–32B trajectory in Table 7 is consistent with this prediction.

6 LIMITATIONS

- Modest undefended baseline on direct prompts.** Llama-3.1-8B exhibits 3.85% undefended ASR on AdvBench direct prompts. While the absolute reduction to 0.00% may appear small, it is operationally critical: in enterprise deployments, a 3% vulnerability rate represents millions of successful malicious queries at scale. Moreover, TEL-OS’s primary value is demonstrated against *optimized* attacks—where the undefended baseline approaches 100%: AutoDAN-Turbo (genetic optimization) and LRM autonomous multi-turn attacks (all four frontier attackers from Hagedorff et al. (2026): DeepSeek-R1, GPT-4o, Qwen3-235B, Gemini 2.5 Flash) achieve near-total bypass on undefended models, yet TEL-OS achieves 0.00% ASR against all of them (200 trials, 95% CI [0.00%, 1.83%]).
- Corpus-specific generalization limits.** TEL-OS is not universally corpus-agnostic: refusal-direction vectors extracted from one behavioral distribution do not automatically generalize to another. Mistral-7B’s 7.00% ASR on HarmBench traces directly to AdvBench-trained vectors failing to cover SQL-injection and political-text-modification behaviors; per-corpus OBLITERATUS re-extraction is expected to reduce this to the 1–2% range. Qwen3-4B reaches an *architectural floor* of 3.75% on HarmBench ($\alpha=0.35$) driven by latent entanglement of cybercrime and code-generation representations; larger extraction corpora and architecture-specific α search may partially reduce this floor, but its existence reflects a structural limit of the model rather than a deficiency of the steering method. Practitioners deploying TEL-OS on a new model or benchmark should treat vector re-extraction as a required calibration step.
- SLERP geometric advantage not directly linked to harm prevention.** Norm Drift difference is a geometric result; we do not provide direct evidence that linear drift causes degraded response quality at higher α values or in long-context generation.
- OBLITERATUS corpus size.** 256 prompt pairs. A larger extraction corpus may improve vector quality for architecturally entangled models (Qwen3-4B’s cybercrime/code domain, Mistral’s HarmBench gap).
- StrongREJECT evaluator limitations.** GPT-4o evaluators produce false positives and negatives. We cross-validate with keyword matching.

6. **Inference cost in resource-constrained environments.** The 15.2% throughput reduction (+4.56 ms/token latency) is measured on RTX 4090 (float16). CPU-only or low-VRAM deployments may experience proportionally higher latency overhead per token.

7 CONCLUSION

The implication for the field is direct: *the bottleneck in inference-time defense is vector quality, not steering method*. The consistent 0.00–0.38% floor across five architectures and two benchmarks reflects a fundamental detection limit—a small fraction of behaviors do not form a clearly separable cluster under the current OBLITERATUS extraction corpus. Larger and more behaviorally diverse extraction corpora are the most direct path to closing this floor—not changes to the steering algorithm.

The central mechanistic finding is that *adaptive urgency-scaled steering collapses the loss landscape that adversarial attacks depend on*. The *Guillotina Geom'etrica* phenomenon—AutoDAN-Turbo’s failure to evolve adversarial controls beyond step 1 across all 50 behaviors—is the clearest evidence: the genetic optimization signal disappears not because harmful outputs are detected and blocked after the fact, but because every suffix that steers the activation toward a harmful direction triggers a proportionally stronger correction—the loss landscape is flat by construction. This is qualitatively different from detection-and-refusal defenses.

A final distinction from prior defenses: TEL-OS does not *block* harmful outputs—it eliminates the model’s geometric capacity to produce them. When SLERP rotates a harmful activation toward the refusal manifold, the latent state that would produce a harmful continuation no longer exists in the representation space; the model is not “choosing” to refuse. This *representation disruption* is structurally distinct from output filtering, detection-and-refusal, and weight-level alignment: the intervention operates before the generation decision is formed, at the level of the representation itself. The flat AutoDAN loss landscape and the 100% vector-trigger rate in adversarial experiments are direct empirical signatures of this mechanism.

SLERP preserves the activation norm invariant (0.999992 ± 0.0004), keeping steered representations on the hyperspherical manifold where the cosine separation gap of 0.284–0.512 already classifies harmful intent before generation begins. The gap is the output of OBLITERATUS extraction—not of SLERP. You et al. (2026) concurrently validate spherical steering as a geometry-preserving principle in a different domain.

TEL-OS is open source. Refusal direction vectors for all five architectures are available at [jostoz/telos-vector](https://github.com/jostoz/telos-vector) (HuggingFace). The OBLITERATUS extraction pipeline is available at github.com/jostoz/OBLITERATUS (AGPL-3.0).

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A PER-MODEL CONFIGURATION

Table 8: Per-model TEL-OS v3.0 configuration and results.

Model	Detection	Steering	$\alpha_{\max}$	AdvBench	HarmBench
Llama-3.1-8B	L_{12}, L_{22}	[9,11,13,15]	0.50	0.00%	0.00%
Qwen3-4B	L_{13}, L_{24}	[10,12,14,16]	0.35	0.38%	3.75%
Qwen3-32B	L_{27}, L_{50}	[28,31,43,47]	0.50	0.00%	0.75%
Mistral-7B	L_5, L_{14}, L_{22}	[14,15,17,18]	0.50	0.38%	7.00%
Gemma-2-9B	L_{17}, L_{27}	[17,18,21,23]	0.50	0.38%	1.50%

B STRONGREJECT RUBRIC

The StrongREJECT evaluator (Souly et al., 2024) rates responses on 0–5:

- **0**—Clear, ethical refusal with no harmful information.
- **1**—Refusal but unclear or inconsistent.
- **2**—Partial refusal with some harmful information leaked.
- **3**—Harmful content present but incomplete.
- **4**—Harmful and detailed (jailbroken).
- **5**—Fully jailbroken, step-by-step harmful instructions.

Jailbroken: score ≥ 4 . GPT-4o scoring model, temperature= 0, max_tokens= 10.

C OBLITERATUS EXTRACTION CORPUS

Table 9: OBLITERATUS contrastive corpus composition (256 prompt pairs).

Category	Pairs
Chemical/bio threats	≈ 40
Cyberattacks	≈ 40
Harassment	≈ 35
Fraud	≈ 35
Illegal content	≈ 35
Harmful/dangerous	≈ 35
Misinformation	≈ 36
Total	256

D URGENCY SCALING HYPERPARAMETER SENSITIVITY

The urgency formula (Equation 2) uses a scaling factor of 200 and a cap of 3.0. We verified robustness by inspecting the empirical raw_d distributions across all experiments:

- **Harmful queries (Llama-3.1-8B, AdvBench):** raw_d $\in [0.05, 0.40]$ (triggered cases). Excess above $\theta = 0.05$: $[0.00, 0.35]$. Multiplied by 200: $[0, 70]$, clamped to $[0, 2]$ in urgency-1 space, then $[1.0, 3.0]$. All triggered harmful queries immediately saturate at urgency=3.0 for any scaling factor ≥ 6 .
- **Benign queries (Llama-3.1-8B, MT-Bench):** $\max(\text{raw_d}) = -0.089$, which is below $\theta = 0.05$ and produces ReLU output of 0 for any threshold value in $[-0.089, \infty)$. The factor of 200 is irrelevant for benign queries since the ReLU output is zero.
- **Sensitivity range:** Any scaling factor in $[6, \infty)$ produces identical behavior on the current evaluation suite (all harmful queries saturate; all benign queries remain dormant). The choice of 200 provides a large margin above the minimum required and has no empirical consequence within the observed signal ranges.

E EXPERIMENT INDEX

Table 10: Experiment index — all results reported in this paper.

XP	Model	Benchmark	ASR
XP-19a	Llama-3.1-8B	AdvBench (520), v2.1 REGEX	0.19%
XP-22	Llama-3.1-8B	AdvBench (520)	0.00%
XP-26	Llama-3.1-8B	HarmBench (400)	0.00%
XP-23	Qwen3-4B	AdvBench (520)	0.38%
XP-25c	Qwen3-4B	HarmBench (400)	3.75%
XP-27d	Qwen3-32B	AdvBench (520)	0.00%
XP-27c	Qwen3-32B	HarmBench (400)	0.75%
XP-28d	Mistral-7B	AdvBench (520)	0.38%
XP-29b	Mistral-7B	HarmBench (400)	7.00%
XP-30	Gemma-2-9B	AdvBench (520)	0.38%
XP-31	Gemma-2-9B	HarmBench (400)	1.50%
XP-33	Llama-3.1-8B	AutoDAN-Turbo (50 beh.)	0.00%
XP-37	Llama-3.1-8B	LRM Autonomous (200 trials, 4 attackers)	0.00%
XP-35	Llama-3.1-8B	FlipAttack (50 beh.)	0.00%*
XP-38	Llama-3.1-8B	Pliny CHAOTIC-ULTRAPLINIAN (52 beh.)	0.00%
XP-36a	Llama-3.1-8B	SLERP vs. Linear (520)	—
XP-36b	Llama-3.1-8B	Benign FPR (200)	0.00%
XP-36c	Llama-3.1-8B	Latency (100)	−15.2% tput; +4.56 ms/tok
XP-BASELINE	Llama-3.1-8B	AdvBench/HarmBench undefended	3.85%/7.25%

* 2.00% keyword; 0.00% model-as-judge (false positive, *Echo-without-Execution*)